بسم الله الرحمن الرحيم

Physics II (EBA1104 & BA114)

Section

Specific Heat and Calorimetry

1. A 50 g sample of copper is at $25°C$. If 1200 J of energy is added to it by heat, what is the final temperature of the copper? **[C_{copper} = 387 J/kg °C]**.

$m_{Cu} = 50\ g = 50\times10^{-3}\ Kg$ $\quad T_i = 25\ °C$ $\quad Q = 1200\ J$

$$Q = m\,c\,\Delta T \longrightarrow Q = m_{cu}\,C_{cu}\,(T_f - T_i)$$

$$1200 = 50\times10^{-3}\times387\times(T_f - 25)$$

$$T_f = 87\ °C$$

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2. An aluminum cup of mass 200 g contains 800 g of water in thermal equilibrium at $80°C$. The combination of cup and water is cooled uniformly so that the temperature decreases by $1.5°C$ per minute. At what rate is energy being removed by heat? Express your answer in watts.
[C_{water} = 4186 J/kg °C, $C_{Aluminum}$ = 900 J/kg °C].

$m_{Al} = 200\ g$	$m_w = 800\ g$
$T_{Al} = 80\ °C$	$T_w = 80\ °C$

$\Delta T = -1.5\ °C$

$$Q = m\,c\,\Delta T$$

$$Q = \left(m_{Al}\,C_{Al} + m_w\,C_w\right)\Delta T$$

$$Q = \left(200\times10^{-3}\times900 + 800\times10^{-3}\times4186\right)\times(-1.5)$$

$$Q = -5293.2\ J$$

$$P = \frac{Q}{t} = \frac{5293.2}{60} = 88.22\ \text{Watt}$$

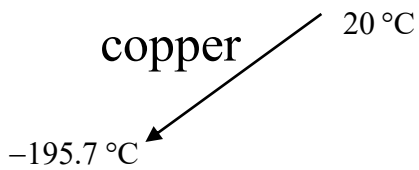
Latent Heat

4. A 1 kg block of copper at 20.0°C is dropped into a large vessel of liquid nitrogen at 77.3 K. How many kilograms of nitrogen boil away by the time the copper reaches 77.3 K ? (The specific heat of copper is 0.0920 $cal/g.°C$, and the latent heat of vaporization of nitrogen is 48.0 cal/g.)

copper

$T_{Cu} = 20\ °C$

$m_{Cu} = 1\ Kg$



Nitrogen

$T_N = 77.3\ K = -195.7\ °C$

$m_N = ??$

$$T_K = T_C + 273$$

$$77.3 = T_f + 273$$

$$T_f = -195.7\ °C$$

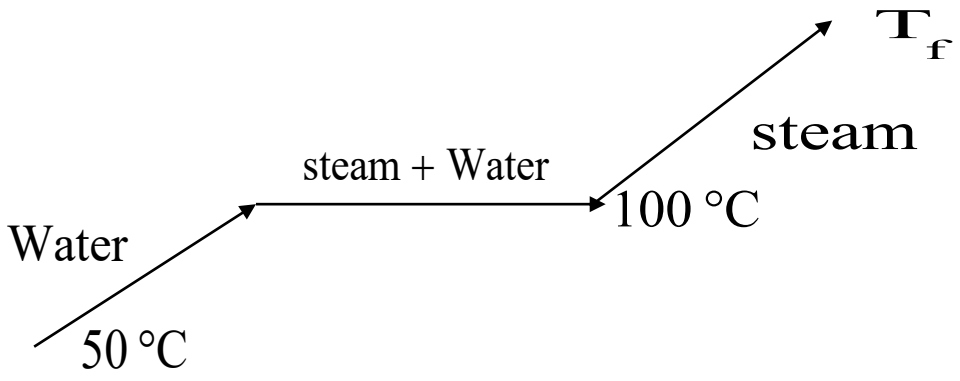
$$Q_{gain} + Q_{loss} = 0$$

$$m_N L_v + (m\,c\,\Delta T)_{cu} = 0$$

$$m_N \times 48 + 1\times10^3 \times 0.092(-195.7 - 20) = 0$$

$$m_N = 4.1\times10^2\ g$$

5. Find the amount of heat required to convert 20 g of water at 50 °C to steam at 120 °C.
[C_{water} = 4186 J/kg °C , C_{steam} = 2010 J/kg °C ,L_v = 2.26 × 10^6 J/Kg].



$$Q = m_w C_w (T_f - T_i) + m_w L_V + m_w C_S (T_f - T_i)$$

$$Q = 20\times10^{-3}\times4186\times(100-50)+20\times10^{-3}\times2.26\times10^{6}+20\times10^{-3}\times2010\times(120-100)=50190\ \text{J}$$

THANKS